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Research Project

Uplift Modeling in Recommender Systems

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Annotation

The topic of recommender systems (RecSys) is popular nowadays as it allows businesses to customize the experience of each of their users individually. Realisation with standard approaches like content- or collaborative-based filtering allows to identify the suggestions appropriate in the context of relevance. However, such selected proposals are subject to various biases due to the fact that algorithms overlook the causality effect. A set of techniques that allow to estimate how impactful recommendations are on user's actions is called uplift modeling. The current paper presents a research aimed at the study of topic's nuances. Through the implemetation of PropCare and DLCE algorithms the analysis of the relevance factor of formed uplift recommendations is performed. In particular, how it relates to algorithm's performance and whether its effect can be incorporated.

Keywords

ML, DL, RecSys, Uplift Modeling

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1 Introduction

Recommender systems (RecSys) have become an important part of the modern data-driven approach to service personalisation enhancement. The general formulation of the task is to recognize the patterns of user behavior and preferences. Then the algorithm selects the most appropriate user-item pair from the pool of available variants and provides a personalized recommendation. The interactions are updated through the feedback loop, which enhances the overall quality of algorithm's future proposals.

The implementation is usually done through either collaborative filtering or content-based methods. The former assumes that there exists a positive correlation in interactions between users with similar features. One of the most popular approaches from this field is the Matrix Factorization (MF) algorithm [13]. The latter mentioned approach is aimed at the prediction and maximization of the click-through rate by providing a user with similar items [26, 28].

Each method discussed overlooks an important part that correlation does not imply causation. The statement refers to the fact that the activity with an item (whether it is click, rating, purchase, etc.) might have been realised even without the algorithm's proposal. Therefore, the introduced potential bias [9] tends to weaken the quality of results produced. Thus, a discussion of uplift modeling, which is a causal approach to RecSys, is needed to estimate the impact recommendations have on user actions.

Through the incorporation of causality into the process, uplift modeling helps to achieve several potential benefits. To begin with, the pesonalization is enhanced as the recommendations are now characterised by greater user dedicated. In addition, the increase in efficiency is featured as the computational resources are aimed at the results with most impact. Finally, in order to capture the causality effect of recommendations, the new metrics are introduced to evaluate the success of different approaches in this field.

The research on the topic was performed as a part of the group work which was to collectively study methods and challenges related to uplift. The current paper is aimed at the accomplishment of two goals with reference to which further theory and material will be presented. First and foremost, through the implementation of selected approaches and their various modifications perform simulations to study the differences and similarities in uplift and relevance maximisation recommendations. Furhtemore, examine the opportunities to balance the tradeoff of uplift-relevance relation identified.

The current paper is structured in the following way. In the following Section (2) the overview of existing methods for uplift modeling is presented with the justification of selected

methods. Then in Section (3) the mathematical formulation of the problem is described as well as the modifications proposed with the specifity of formalised goals. After this in Section (4) the description of used data and conducted experiments is presented. The Section (5) serves as the closing one formalizing the conclusions from the results obtained at the current stage and providing ideas for future development.

2 Literature review

The research on causality-aware RecSys emerged from the ideas of uplift modeling applications. The key advantage of causal effect account is that it helps to decrease the effect of various biases (popularity, noise, etc.) [9]. Most important derivation from it in the scope of the current paper is that it provides with an opportunity to measure the effect produced recommendations have on users.

2.1 Existing approaches

As stated in the Section (1) one of the most renowned approaches for RecSys is MF [13]. Nevertheless, its most basic version assumes that the observed data is the true preference of a user and introduces selection bias [25].

The growing interest in causal RecSys [27] has provided with numerous approaches each based on different assumptions trying to eliminate potential biases. For example, one of the algorithms called CausE [2] builds upon the ideas of MF but aims to eliminate confounding factors by introducing two training datasets of large and small. The former contains observations obtained from some selected recommender policy and the latter from unbiased algorithm (e.g. random). Leveraging the multi-stage learning the approach leads to the separation of environment's effects on user recommendations.

Another approach named CausalNeighborhood [23] provides with an alternative assumptions and uses the idea of k-nearest neighbors (kNN) to calculate the similarity of interactions between users and items. The produced ranked lists are then evaluated in terms of causal effect and final recommendations are formed.

For the mentioned approaches to work the exposure data is necessary. One of the possible methods for its estimation is CJBPR [34]. This unbiased approach is aimed at the learning of user-item relevance and propensity. However, its own training requires the full data collection containing information about recommendation provision.

2.2 Selected approaches

As stated in the Subsection (2.2) the most prominent challenge to exist is the provision of algorithms with exposure data and its estimation. For this reason the current paper focuses on the PropCare approach [16]. Its main advantage is that it estimates the propensity utilising user-item interactions only. In addition, alike CJBPR it also allows to estimate the probability of relevance of an item for a user. This information will be used for modifications described in Section (3) with the motivation and results presented in Section (4).

Considering the causal approach for recommendations generation, the DLCE [24] algorithm was selected. Unlike CausE that requires two datasets and correlation influence present in CausalNeighborhood, it continues to elevate the ideas of MF calculating the causal effect directly by leveraging estimation propositions of [10, 20, 24]. Besides, the comparison of performance of DLCE to previously discussed approaches is done as a separate part of the group work.

2.3 Related works

With the references to selected approaches described in Subsection (2.2) the existing related papers were analysed to verify the necessity and uniqueness of the identified goal to study the relevance factor of produced recommedations.

Considering the methodologies linked to PropCare, several can be discussed. One of the papers discusses an approach named "CausalX" [14] aimed at the use of LLM for increased transparency and debiasing of the recommendations. Despite the causal consistency enforced by the incorporation of explanations, this methodology does not directly estimate the exposure data and requires substantial computational resources. Another technique described in [11] intorduces a multifactorial bias correction with alternated gradient descent. The handling of more complex interactions comes with the need for explicit ratings data, which is usually unavailable in many real-life cases. Last related research discussed in [1] is aimed at stochastic session modeling to capture temporal dynamics. Nevertheless, its narrow scope causes dependence on time lengths and limited scalability.

The collection of related researches for DLCE is more extensive and can be classified into a number of discussed fields. The first one is connected to the topic of causal inference techniques [17, 30, 31] with various studies in modeling of causal graphs [32] to capture more complex relationships or exploration of more consistent estimators [21]. The increase in robustness is obtained with increased complexity in implementation and computation and greater reliance on data specifity. The goal of alternative debiasing is discussed by [12, 22] or solved through neural

networks leverage [7] which both risk overfitting. Studies in unlearning biased user preferences [8, 33] or increase of algorithm's fairness in recommendations proposal [4, 6] have narrow causal scope. Other researches like targeting specific niches (e.g. news [18], cloud services [5]), using knowledge distillation techniques [3] and simulating counterfactual scenarios [29] are prone to have dependencies from contextual data, loss of causal interpretability and training instabilities. All the mentioned works highlight the DLCE benefit as an introductory universal framework for uplift-specific challenges exploration.

3 Problem statement

3.1 Basic notations

For the purpose of inutitive explanation provision alongside the mathematical formulation of the selected approaches described in Subsection (2.2), Table 3.1 contains basic notations that will be used to provide further interpretations.

Notation	Definition
$u \in \{1, ..., U\}$	specific user from a set of users
X_u	feature vector of user u
$i \in \{1, ..., I\}$	specific item from a set of items
X_i	feature vector of item i
pop_i	popularity of item i
X_F	feature vector of frequency interactions (users-items)
$X_F(u), X_F(i)$	feature vector of frequency interactions (user-items or users-item)
$Z_{u,i} \in \{0, 1\}$	indication whether user u was recommended with item i / belongingness to the treatment group
$p_{u,i} \in [0, 1]$	probability that user u was recommended with item i / propensity
$Y^T_{u,i} \in \{0, 1\}$	outcome from the interaction of user u with item i under treatment group
$Y^C_{u,i} \in \{0, 1\}$	outcome from the interaction of user u with item i under control group
$Y_{u,i}$	outcome (rating, purchase, etc.) from the interaction of user u with item i $(Z_{u,i}Y^T_{u,i} + (1 - Z_{u,i})Y^C_{u,i})$
$R_{u,i} \in \{0, 1\}$	indication wherher item i is relevant for user u
$r_{u,i} \in \{0, 1\}$	probability that item i is relevant for user u

Table 3.1: Basic notations used in the paper

Refering to the introduced notations, the first definition that is necessary to be presented is how the causal effect is formulated and measured. It is defined as [20]

$$\tau_{u,i} = Y^T_{u,i} - Y^C_{u,i}. \tag{1}$$

Due to the binary nature of outcomes $\tau \in \{-1, 0, 1\}$. The trivial case is described by the $\tau = 0$ concluding that with or without the recommendation the item is not appealing for the user. In the case of a positive effect ($\tau = 1$) models' suggestion boosted user's interest in the item which wouldn't be possible without it. The opposite situation of $\tau = -1$ suggests that the recommendation prevented user-item interaction appearance. However, the counterfactual nature of causal effect prevents from computing it directly. For this reason the inverse propensity score (IPS) estimator [10, 24] is introduced defined as

$$\hat{\tau}_{u,i} = \frac{Z_{u,i}Y_{u,i}}{p_{u,i}} - \frac{(1 - Z_{u,i})Y_{u,i}}{(1 - p_{u,i})}, \tag{2}$$

where the first and second term are weighted to balance the distribution of underrepresented treatment assignments thus eliminating the bias introduced.

Another important relation [19] to be formalised and used in the definition for PropCare method is defined by the following equation

$$y_{u,i} = p_{u,i} r_{u,i}, \tag{3}$$

where $y_{u,i} = P(Y_{u,i} = 1)$ is the probability of interaction between the user u and item i and $r_{u,i} = P(R_{u,i} = 1)$ is the probability that item i is relevant for user u.

3.2 PropCare

For the convenience of step-by-step explanation let's formalize the mathematical problem for the PropCare algorithm first. Its final loss function is constructed with several separate ones. The fisrt component is defined as

$$\mathcal{L}_{nave} = -Y_{u,i} \log(f_p(x_{u,i}; \theta_p) f_r(x_{u,i}; \theta_r)) - (1 - Y_{u,i}) \log(1 - f_p(x_{u,i}; \theta_p) f_r(x_{u,i}; \theta_r)), \tag{4}$$

where $x_{u,i} = f_e(X_u, X_i; \theta_e)$ is the user-item embedding; f_* is trainable MLP with corresponding vector of weights θ_* for either user, propensity or relevance embedding. The intuition behind this term is to reflect the binary cross-entropy relation with the eq. (3). However, the multiplication

of propensity and relevance respective output embeddings prevents from differentiation of their individual effects. Thus, the validated approach of popularity introduction [16] helps to formulate the second term defined as following

$$\mathcal{L}_{pop} = -\kappa_{u,i,j} \log(\sigma(sgn_{i,j} \cdot (f_p(x_{u,i}) - f_p(x_{u,j}))) + \sigma(sgn_{i,j} \cdot (f_r(x_{u,i}) - f_r(x_{u,j})))), \tag{5}$$

where $\kappa_{u,i,j} = e^{\psi(y_{u,i} - y_{u,j})}$ for hyperparameter $\psi < 0$ is a weighting function, σ is the sigmoid activation function, $sgn_{i,j} \in \{-1, 1\}$ is the sign of $(pop_i - pop_j)$. The two functions are used to produce the final form of PropCare loss defined as

$$\mathcal{L}_{PropCare} = \sum_{u,i,j} (\mathcal{L}_{nave} + \lambda \mathcal{L}_{pop}) + \mu KL(Q || Beta(\alpha, \beta)), \tag{6}$$

where $KL(\cdot||\cdot)$ is the Kullback-Leibler divergence between the distribution of estimated $\hat{p}_{u,i}$ retreived from Q and the beta distribution with hyperparameters α and β; λ and μ are hyperparameters balancing the influence of popularity and regularization. The loss function is minimized to obtain the estimated propensities $\hat{p}_{u,i}$ and predict $\hat{Z}_{u,i}$ through $Norm(\hat{p}_{u,i}) \geq \epsilon$ with ϵ being a threshold parameter and $Norm$ a $Z-$score normalization function.

Considering the modifications introduced to the final loss of PropCare represented by the eq.(6), it was transformed and formalised as

$$\mathcal{L}^{Mod}_{PropCare} = \mathcal{L}_{PropCare} + \lambda_2 \mathcal{L}_{prop} + \lambda_3 \mathcal{L}_{rel} + \nu KL(W || Beta(\alpha, \beta)), \tag{7}$$

where $\mathcal{L}_{prop}, \mathcal{L}_{rel}$ are binary cross-entropy functions for greater separation of influence derived for more precise estimation of propensity and relevance respectively (these two additional terms were realised and incorporated into the discussed modification by another member of the group work); $KL(\cdot||\cdot)$ is the Kullback-Leibler divergence between the distribution of estimated $\hat{r}_{u,i}$ retreived from W and the beta distribution with hyperparameters α and β; $\lambda_2, \lambda_3, \nu$ are hyperparameters controlling the influence of different terms. The addition of the second KL divergence regularization term was done in order to fight the distribution of relevance obtained from the realisation of the initial PropCare algorithm which resembled a distribution with values concentrated either around 0 or 1. The reasoning and results obtained from training of this algorithm are presented in Subsection (4.2).

3.3 DLCE

After the propensities are trained and exposure data is estimated it is passed to the DLCE algorithm to produce recommendations. For each pair of item i and user u the score is estimated defined as

$$\hat{s}_{u,i} = f_M(X_u, X_i; \theta_M), \tag{8}$$

where $f_M(*)$ is a trainable MLP. Items are then sorted by $\hat{s}_{u,i}$ for each user and ranked producing a vector $r(\hat{s}_{u,i})$. From eq. (8) the average metric function is derived presented as

$$R(M) = \frac{1}{UI} \sum_u \sum_i \lambda(r(\hat{s}_{u,i}))\tau_{u,i}, \tag{9}$$

where $\lambda(r(\hat{s}_{u,i})) = \lambda_{u,i} = -r(\hat{s}_{u,i})$ is a weighting function and M denotes model. From it the empirical risk minimization (ERM) function becomes $argmin_M(-R(M))$ containing the counterfactual parameter τ. By substituting it with IPS estimation from eq. (2) and introducing capping for propensity deviation constraint [24], the capped IPS version is derived as

$$R^{CIPS}(M) = \lambda_{u,i}(\frac{Z_{u,i}Y_{u,i}}{\max(p_{u,i}, \chi^T)} - \frac{(1 - Z_{u,i})Y_{u,i}}{\max((1 - p_{u,i}), \chi^C)}) = \lambda_{u,i}\tilde{\tau}_{u,i}, \tag{10}$$

where χ^T, χ^C are capping parameters bounding the denominators.

The final step is to estimate the weightning function $\lambda_{u,i}$, otherwise called as causal average rank (CAR). To perform it the local loss is introduced as

$$\delta_{u,i} = -\lambda_{u,i}\tilde{\tau}_{u,i} = r(\hat{s}_{u,i})\tilde{\tau}_{u,i}, \tag{11}$$

where $r(\hat{s}_{u,i}) = (1+\sum_{j\neq i} \mathbf{1}(\hat{s}_{u,j} > \hat{s}_{u,i}))$. By introducing the approximation of the non-differentiable indicator function [15] and applying some transformational changes the final loss is describes as

$$\begin{aligned}
\delta_{u,i} &= r(\hat{s}_{u,i})\tilde{\tau}_{u,i} \\
&= r(\hat{s}_{u,i})\tilde{Y}^T_{u,i} - (I - r(\hat{s}_{u,i}))\tilde{Y}^C_{u,i} - I\tilde{Y}^C_{u,i} \\
&\leq (\sum_{j\neq i} \log(1 + \exp(-\omega\hat{s}_{u,i,j})))\frac{Z_{u,i}Y_{u,i}}{\max(p_{u,i}, \chi^T)} - \\
&\quad - (\sum_{j\neq i} \log(1 + \exp(\omega\hat{s}_{u,i,j})))\frac{(1 - Z_{u,i})Y_{u,i}}{\max((1 - p_{u,i}), \chi^C)} \\
&= \mathcal{L}_{DLCE},
\end{aligned} \tag{12}$$

where $I = 1 + \sum_{j\neq i} \mathbf{1}(\hat{s}_{u,j} > \hat{s}_{u,i}) + \sum_{j\neq i} \mathbf{1}(\hat{s}_{u,i} > \hat{s}_{u,j})$; $\hat{s}_{u,i,j} = \hat{s}_{u,i} - \hat{s}_{u,j}$; ω is a hyperparameter

for approximation. The obtained loss function is then used to produce the ranking and provide recommendations for the user based on it.

Considering the modifications introduced to the DLCE, there were two major transformations applied. The first one is the altered loss function represented by the eq.(12) which was formalised as

$$\begin{aligned}\mathcal{L}_{DLCE}^{Mod} &= \hat{r}(\hat{s}_{u,i})\tilde{\tau}_{u,i}^{'} \\ &= \hat{r}(\hat{s}_{u,i})(\tilde{\tau}_{u,i} + \phi\tilde{\tau}_{u,i}^{rel}) \\ &= \hat{r}(\hat{s}_{u,i})(\tilde{\tau}_{u,i} + \phi(\frac{R_{u,i}Y_{u,i}}{\max(r_{u,i}, \chi^T)} - \frac{(1-R_{u,i})Y_{u,i}}{\max((1-r_{u,i}), \chi^C)})),\end{aligned} \tag{13}$$

where $\tilde{\tau}_{u,i}^{rel}$ measures ITE of item i relevance on the user u and ϕ regulates the effect it has on the total loss. The second modification done with the rank prediction of recommendation suggests to incorporate the frequency of user-item positive interactions based on history. Three different variants of such transformations are represented as

$$\begin{aligned}\hat{s}_{u,i}^{F} &= f_M(X_u, X_i; \theta_M) + X_F, \\ \hat{s}_{u,i}^{FU} &= f_M(X_u + X_F(u), X_i; \theta_M), \\ \hat{s}_{u,i}^{FI} &= f_M(X_u, X_i + X_F(i); \theta_M) + X_F\end{aligned} \tag{14}$$

The results obtained from training of this algorithm and its modifications are presented and discussed in Subsection (4.2).

4 Experiment setup and results

4.1 Setup description

For the evaluation of PropCare and DLCE of their base and modified versions two datasets were used. The first one is DunnHumby "The Complete Journey"[1]. The data contains information about purchases of 2500 users and promotional logs at a physical retailer for 93 weeks. Information is used to generate synthetic data of two types [23, 24]. The first version named $DH_{original}$ utilises the empirical estimate of propensity dividing the number of times a user bought an item during the weekly mailer ($\sum_t \mathbf{1}(Y_{u,i,t} = 1)$) that recommended the particular item by the total number

[1] https://www.dunnhumby.com/careers/engineering/sourcefiles

of users purchases. The second $DH_{personalised}$ was generated using the setting as following

$$p_{u,i} = \min(1, \alpha(\frac{1}{rank})^{\beta}), \tag{15}$$

where $rank$ is the order in which item for user was placed in terms of purchase probability $P(Y_{u,i} = 1)$. In addition to the initial preprocessing during the research the titles of items and their average prices were retrieved that were lost in the beginning.

The data was generated both for one week [24] and for ten weeks [23]. The results presented in Subsection (4.2) in will be displayed only for the former generation. However, for the frequency information retrieval, the latter dataset was used as well. Also, the division of products based on category was selected. This was done due to the fact that in order to evaluate the results and investigate relevance influence the data requiring less training time was chosen.

The second dataset is based on "FINN.no Slate Dataset for Recommender Systems"[2]. The dataset contains information about users interaction with marketplace products both based on their own search or recommendation. Such variant was chosen to mitigate the effect of synthetic generation and make possible to compare different methods on real information. However, much smaller size of the dataset made must be taken into account and will be addressed further.

The basic princpiples of the training that comprises both exposure data estimation and recommendation generation is described by the Algorithm 1. The base realisation of DLCE was taken from the files that are attached to [24]. The base realisation of PropCare algorithm was taken from open GitHub repository[3] performed by Zhongzhou Liu. The code was modified and corrected with some inaccuracies relating to the description in [16] and overall mistakes noticed. In addition, some of the hyperparameters were changed. The code is stored at the GitHub repository[4].

Two sets of metrics were selected for evaluation. The first one consists of recall, precision and NDCG versions calculated for the first top 10 ranked generated suggestions:

$$NDCG@k = \frac{DCG@k}{IDCG@k} = \frac{\sum_u \sum_{i=1}^{k_{rank}} \frac{Y_{u,i}}{\log_2 i + 1}}{\sum_u \sum_{i=1}^{k_{ideal}} \frac{Y_{u,i}}{\log_2 i + 1}},$$

$$Recall@k = \frac{\sum_u \sum_{i=1}^{k} Y_{u,i}}{\sum_u \sum_i Y_{u,i}},$$

$$Precision@k = \frac{\sum_u \sum_{i=1}^{k} Y_{u,i}}{k}.$$

[2] https://github.com/finn-no/recsys_slates_dataset
[3] https://github.com/mediumboat/PropCare/tree/main
[4] https://github.com/RomanGorelsky/Uplift-Modelling-in-Recommender-Systems

Such metrics were chosen as they measure standard performance without uplift. The causal effect is evaluated differently for two datasets, as only DunnHumby possesses the semi-synthesized causal effect values. They are formalised as following:

$$CP@k = \frac{1}{U}\sum_u \sum_i \frac{\mathbf{1}(rank_u(\hat{s}_{u,i} \leq k))\tau_{u,i}}{k} \qquad Uplift@k = \frac{\sum_i Y_i^T}{k} - \frac{\sum_i Y_i^C}{k}$$

$$CDCG = \frac{1}{U}\sum_u \sum_i \frac{\tau_{u,i}}{\log_2(1 + rank_u(\hat{s}_{u,i}))} \qquad Weighted\ Average\ Uplift = \frac{\sum_k n_k^T\, Uplift@k}{\sum_k n_k^T}.$$

The left part denotes metrics used with DunnHumby aimed at measuring the reward coming from the ranking of items in the first top-k list of recommendations that have positive effect on user-item interaction. The right ones help to evaluate causal performance in the Finn.no setting.

To conclude, the comparison in metrics of Algorithm 1 and its modifications was done with the recommendation ranking based on the probability of relevance produced by PropCare (row "Relevance (R)") and two baselines ranking based on popularity and its modified version named personal popularity calculated the train dataset (rows "Popularity (P)" and "Personal Popular (PP)"). The relevance sorting is not utilised in Finn.no setting as it doesn't require to estimate propensity values and is rather used to verify the conclusions made based on results from DunnHumby.

4.2 Results description

The selected approaches were realised in their initial form and their performance was firstly measured on two semi-synthesised DunnHumby datasets. Tables 4.1 and 4.2 depict the results obtained after training both PropCare and DLCE 3 consecutive times.

Discussing the $DH_{original}$ setting with emprical estimation of the causal effect, several things can be noticed. First of all, the performance of non-modified version of DLCE is worse than the baseline of popularity ranking. The assumption for this part is that such an outcome is dictated by the approach of data generation in $DH_{original}$. In addition, a clear conclusion that personal popularity model outperformed all other rankings. Nevertheless, it can also be noticed that DLCE ($\hat{s}^F_{u,i}$) version results are quite similar (even better for CP@10) and exceed the basic popularity ones. Finally, relevance sorting displays worse results among all approaches.

Table 4.1: Metrics with $\mathcal{L}_{PropCare}$ on $DH_{original}$ (Mean ± Std. Error)

Method	**NDCG@10**	**Recall@10**	**Precision@10**
Relevance (R)	0.0014 ± 0.0005	0.0010 ± 0.0003	0.0013 ± 0.0005
Popularity (P)	0.2223 ± 0.0	0.1202 ± 0.0	0.1755 ± 0.0
Personal Popular (PP)	$\mathbf{0.2477 \pm 0.0}$	$\mathbf{0.1395 \pm 0.0}$	$\mathbf{0.2049 \pm 0.0}$
DLCE ($\hat{s}_{u,i}$)	0.1952 ± 0.0046	0.1106 ± 0.0018	0.1631 ± 0.0024
DLCE ($\hat{s}^{F}_{u,i}$)	0.2434 ± 0.0013	0.1365 ± 0.0010	0.2011 ± 0.0016
DLCE ($\hat{s}^{FI}_{u,i}$)	0.1148 ± 0.0215	0.0678 ± 0.0113	0.0981 ± 0.0175
DLCE ($\hat{s}^{FU}_{u,i}$)	0.0960 ± 0.0183	0.0545 ± 0.0100	0.0826 ± 0.0149

Method	**CP@10**	**CP@100**	**CDCG**
Relevance (R)	0.0022 ± 0.0003	0.0024 ± 0.0003	0.6677 ± 0.0021
Popularity (P)	0.0293 ± 0.0	0.0155 ± 0.0	0.9251 ± 0.0
Personal Popular (PP)	0.0315 ± 0.0	$\mathbf{0.0164 \pm 0.0}$	$\mathbf{0.9396 \pm 0.0}$
DLCE ($\hat{s}_{u,i}$)	0.0284 ± 0.0014	0.0147 ± 0.0006	0.8928 ± 0.0097
DLCE ($\hat{s}^{F}_{u,i}$)	$\mathbf{0.0349 \pm 0.0001}$	0.0158 ± 0.0002	0.9187 ± 0.0069
DLCE ($\hat{s}^{FI}_{u,i}$)	0.0167 ± 0.0018	0.0112 ± 0.0005	0.8185 ± 0.0123
DLCE ($\hat{s}^{FU}_{u,i}$)	0.0155 ± 0.0029	0.0108 ± 0.0009	0.8155 ± 0.0158

For the $DH_{personalised}$ case the situation differs quite substantially. To begin with, the statistics for base DLCE version are enhanced compared to popularity in the case of causal metrics. Furthermore, personal popularity approach once again dominates across all metrics. But for this case the values for DLCE ($\hat{s}^{F}_{u,i}$) differ quite a lot. Considering relevance scores, they again remain low.

Table 4.2: Metrics with $\mathcal{L}_{PropCare}$ on $DH_{personalised}$ (Mean ± Std. Error)

Method	**NDCG@10**	**Recall@10**	**Prec@10**
Relevance (R)	0.0012 ± 0.0004	0.0008 ± 0.0001	0.0012 ± 0.0002
Popularity (P)	0.2316 ± 0.0	0.1150 ± 0.0	0.1870 ± 0.0
Personal Popular (PP)	$\mathbf{0.3431 \pm 0.0}$	$\mathbf{0.1826 \pm 0.0}$	$\mathbf{0.2938 \pm 0.0}$
DLCE ($\hat{s}_{u,i}$)	0.1816 ± 0.0004	0.1052 ± 0.0010	0.1699 ± 0.0012
DLCE ($\hat{s}^{F}_{u,i}$)	0.2892 ± 0.0014	0.1495 ± 0.0008	0.2449 ± 0.0013
DLCE ($\hat{s}^{FI}_{u,i}$)	0.1400 ± 0.0270	0.0764 ± 0.0138	0.1228 ± 0.0225
DLCE ($\hat{s}^{FU}_{u,i}$)	0.1602 ± 0.0026	0.0819 ± 0.0046	0.1361 ± 0.0066

Method	**CP@10**	**CP@100**	**CDCG**
Relevance (R)	0.0015 ± 0.0002	0.0019 ± 0.0002	0.6128 ± 0.0018
Popularity (P)	0.0275 ± 0.0	0.0157 ± 0.0	0.8646 ± 0.0
Personal Popular (PP)	$\mathbf{0.1387 \pm 0.0}$	$\mathbf{0.0392 \pm 0.0}$	$\mathbf{1.4728 \pm 0.0}$
DLCE ($\hat{s}_{u,i}$)	0.0569 ± 0.0003	0.0172 ± 0.0002	0.9438 ± 0.0018
DLCE ($\hat{s}^{F}_{u,i}$)	0.0959 ± 0.0003	0.0271 ± 0.0001	1.2195 ± 0.0003
DLCE ($\hat{s}^{FI}_{u,i}$)	0.0407 ± 0.0113	0.0127 ± 0.0032	0.8616 ± 0.0701
DLCE ($\hat{s}^{FU}_{u,i}$)	0.0657 ± 0.0090	0.0144 ± 0.0012	0.9889 ± 0.0340

For the purpose of uplift-relevance relationship study, statistics for all discussed cases were collected to investigate changing dynamics and improvements. To be clear, the comparison was done between the DLCE rankings from ($\hat{s}_{u,i}$), DLCE ($\hat{s}^F_{u,i}$) and personal popular prediction approaches and the relevance estimates received by the PropCare. Thus, results are prone to deviate and complex study with multiple metrics is needed.

For this reason the ranking approaches were studied through the analysis of average rank differences, Kendall tau correlation coefficient calculation and top-k overlap scores comparison. In the case of relevance estimates with $\mathcal{L}_{PropCare}$ the statistics are depicted in the Tables 4.3 and 4.4.

Table 4.3: Comparison on $DH_{original}$ with $\mathcal{L}_{PropCare}$

Method	**Avg. Rank Diff.**	**Kendall Tau**	**Top 10 Overlap**	**Top 100 Overlap**
DLCE ($\hat{s}_{u,i}$)	486.4081	-0.0255	0.000000	0.000030
DLCE ($\hat{s}^F_{u,i}$)	495.9260	-0.0256	0.000087	0.000260
Personal Popular (PP)	674.2838	-0.0300	0.001732	0.007371

Table 4.4: Comparison on $DH_{personalised}$ with $\mathcal{L}_{PropCare}$

Method	**Avg. Rank Diff.**	**Kendall Tau**	**Top 10 Overlap**	**Top 100 Overlap**
DLCE ($\hat{s}_{u,i}$)	498.1515	-0.0776	0.000000	0.000013
DLCE ($\hat{s}^F_{u,i}$)	503.2412	-0.0776	0.000087	0.000329
Personal Popular (PP)	667.7532	-0.0782	0.001732	0.009593

For both settings it can be observed the situation that there is negative relationship between all approaches and relevance ranking. This is also supported by the fact that produced recommendation lists overlap quite rarely in first 10 items with only personal popular methodology yielding at least noticeable non-zero results.

After the completion of the base models implementation and their evaluation it was noticed that the distribution of estimated relevance probabilities for recommended items derived from PropCare was subject to concentration either around 0 or 0.8 (as it was scaled by this value). The example of such effect is depicted on the Figure 4.1. Thus, it was decided to make modifications to the basic PropCare loss function in eq. (6) described in Subsection (3.2).

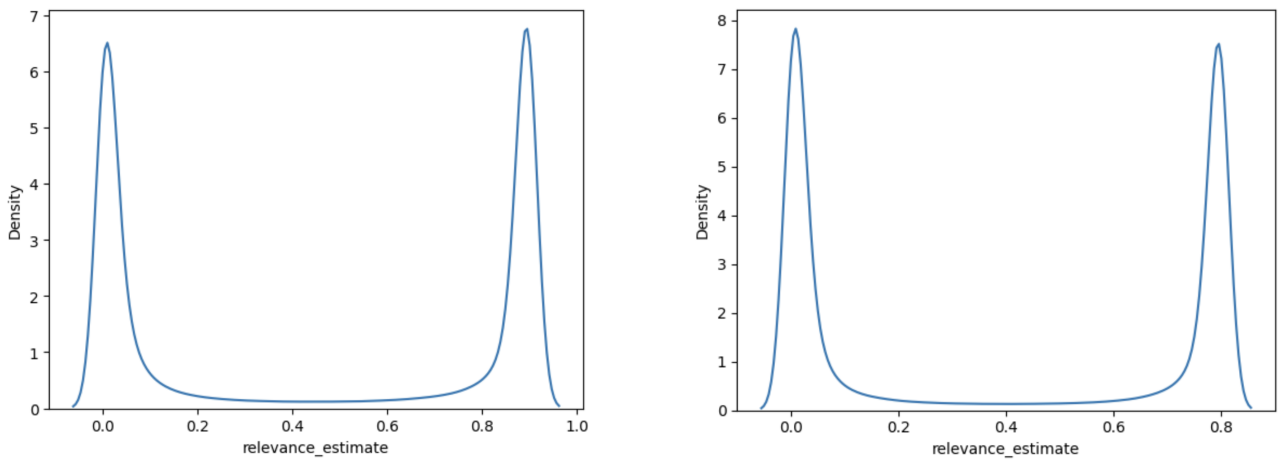

Figure 4.1: Distribution of $\hat{r}_{u,i}$ with $\mathcal{L}_{PropCare}$ for $DH_{original}$ (left) and $DH_{personalised}$ (right)

The introduced terms in eq. (7) are aimed both at greater regularization of parameters and more efficient separation of effects propensity and relevance have on the outcome. Considering the parameters λ_2 and λ_3 were set to 0.1 and ν was set equal to μ. The new simulations were performed with the effect modifications implementation brought on relevance estimation depicted on Figure 4.2. The dispersion of the right-hand peak denotes more accurate and diverse values for $\hat{r}_{u,i}$

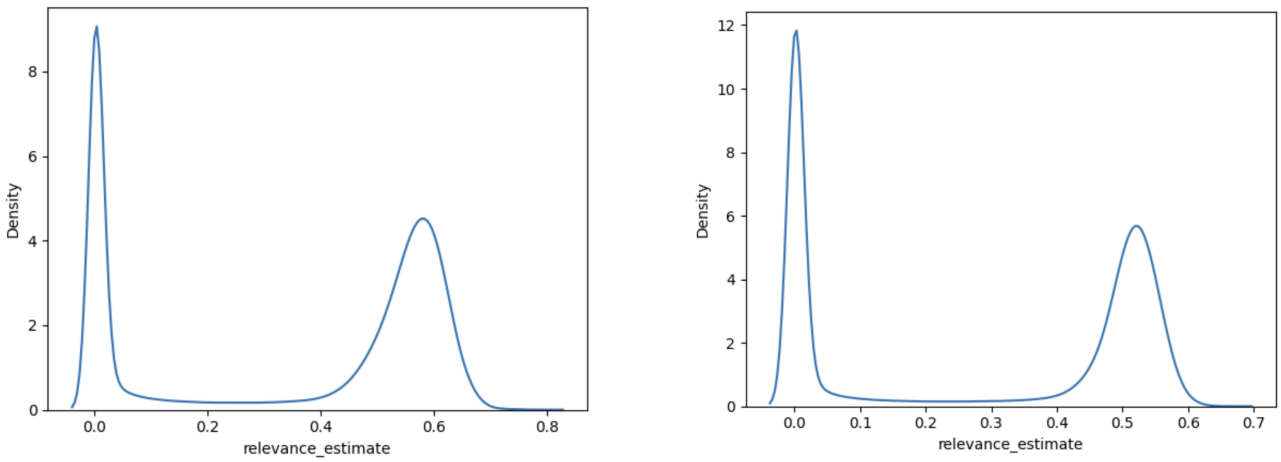

Figure 4.2: Distribution of $\hat{r}_{u,i}$ with $\mathcal{L}^{Mod}_{PropCare}$ for $DH_{original}$ (left) and $DH_{personalised}$ (right)

New simulations performed to study the effect of modifications on metrics for all ranking approaches. The results are contained in Tables 4.5 and 4.6.

Considering the $DH_{original}$ setting, the results are to be analysed from different perspectives. First and foremost, it can be observed that the relevance metrics improved suggesting that $\mathcal{L}^{Mod}_{PropCare}$ tends to produce more accurate estimates. However, it must be mentioned that standard deviation increased as well which affects the stability of algorithm. In the meantime, DLCE results depict changes that are similar to standard deviation recorded in Table 4.1. Thus, the

introduced changes do not negatively alter the statistics in the discussed settings for the models reinforcing their idea.

Table 4.5: Metrics with $\mathcal{L}^{Mod}_{PropCare}$ on $DH_{original}$ (Mean ± Std. Error)

Method	**NDCG@10**	**Recall@10**	**Precision@10**
Relevance (R)	0.0117 ± 0.0008	0.0072 ± 0.0007	0.0109 ± 0.0011
Popularity (P)	0.2223 ± 0.0	0.1202 ± 0.0	0.1755 ± 0.0
Personal Popular (PP)	$\mathbf{0.2477 \pm 0.0}$	$\mathbf{0.1395 \pm 0.0}$	$\mathbf{0.2049 \pm 0.0}$
DLCE ($\hat{s}_{u,i}$)	0.1982 ± 0.0017	0.1097 ± 0.0005	0.1618 ± 0.0008
DLCE ($\hat{s}^{F}_{u,i}$)	0.2455 ± 0.0006	0.1381 ± 0.0004	0.2035 ± 0.0006
DLCE ($\hat{s}^{FI}_{u,i}$)	0.1533 ± 0.0313	0.0887 ± 0.0164	0.1292 ± 0.0243
DLCE ($\hat{s}^{FU}_{u,i}$)	0.1272 ± 0.0476	0.0741 ± 0.0274	0.1112 ± 0.0389

Method	**CP@10**	**CP@100**	**CDCG**
Relevance (R)	0.0130 ± 0.0038	0.0045 ± 0.0006	0.7492 ± 0.0216
Popularity (P)	0.0293 ± 0.0	0.0155 ± 0.0	0.9251 ± 0.0
Personal Popular (PP)	0.0315 ± 0.0	$\mathbf{0.0164 \pm 0.0}$	$\mathbf{0.9396 \pm 0.0}$
DLCE ($\hat{s}_{u,i}$)	0.0289 ± 0.0016	0.0143 ± 0.0007	0.9042 ± 0.0028
DLCE ($\hat{s}^{F}_{u,i}$)	$\mathbf{0.0328 \pm 0.0006}$	0.0161 ± 0.0004	0.9254 ± 0.0048
DLCE ($\hat{s}^{FI}_{u,i}$)	0.0230 ± 0.0022	0.0133 ± 0.0013	0.8644 ± 0.0173
DLCE ($\hat{s}^{FU}_{u,i}$)	0.0197 ± 0.0042	0.0126 ± 0.0015	0.8475 ± 0.0321

For the $DH_{personalised}$ case conclusions are similar for the relevance case. However, it can also be observed that results for DLCE in the causal metrics improved quite significantly with the values of standard ones remaining in acceptable range.

Table 4.6: Metrics with $\mathcal{L}^{Mod}_{PropCare}$ on $DH_{personalised}$ (Mean ± Std. Error)

Method	**NDCG@10**	**Recall@10**	**Precision@10**
Relevance (R)	0.0145 ± 0.0032	0.0090 ± 0.0014	0.0146 ± 0.0022
Popularity (P)	0.2316 ± 0.0	0.1150 ± 0.0	0.1870 ± 0.0
Personal Popular (PP)	$\mathbf{0.3431 \pm 0.0}$	$\mathbf{0.1826 \pm 0.0}$	$\mathbf{0.2938 \pm 0.0}$
DLCE ($\hat{s}_{u,i}$)	0.1800 ± 0.0023	0.1042 ± 0.0010	0.1686 ± 0.0018
DLCE ($\hat{s}^{F}_{u,i}$)	0.2946 ± 0.0028	0.1513 ± 0.0016	0.2480 ± 0.0023
DLCE ($\hat{s}^{FI}_{u,i}$)	0.1582 ± 0.0100	0.0859 ± 0.0053	0.1377 ± 0.0084
DLCE ($\hat{s}^{FU}_{u,i}$)	0.1843 ± 0.0215	0.0990 ± 0.0140	0.1626 ± 0.0221

Method	**CP@10**	**CP@100**	**CDCG**
Relevance (R)	0.0049 ± 0.0024	0.0044 ± 0.0008	0.6613 ± 0.0135
Popularity (P)	0.0275 ± 0.0	0.0157 ± 0.0	0.8646 ± 0.0
Personal Popular (PP)	$\mathbf{0.1387 \pm 0.0}$	$\mathbf{0.0392 \pm 0.0}$	$\mathbf{1.4728 \pm 0.0}$
DLCE ($\hat{s}_{u,i}$)	0.0591 ± 0.0010	0.0185 ± 0.0003	0.9572 ± 0.0048
DLCE ($\hat{s}^{F}_{u,i}$)	0.1030 ± 0.0027	0.0294 ± 0.0003	1.2695 ± 0.0084
DLCE ($\hat{s}^{FI}_{u,i}$)	0.0513 ± 0.0040	0.0165 ± 0.0015	0.9302 ± 0.0284
DLCE ($\hat{s}^{FU}_{u,i}$)	0.0700 ± 0.0086	0.0175 ± 0.0021	1.0161 ± 0.0477

Concluding the results for the simulations with $\mathcal{L}^{Mod}_{PropCare}$, exploration of changes in uplift-relevance relationship is performed. The results contained in Tables 4.7 and 4.8. The statistics show a considerable increase across all metrics and denotes a positive association between relevance and studied approaches rankings.

Table 4.7: Comparison on $DH_{original}$ with $\mathcal{L}^{Mod}_{PropCare}$

Method	**Avg. Rank Diff.**	**Kendall Tau**	**Top 10 Overlap**	**Top 100 Overlap**
DLCE ($\hat{s}_{u,i}$)	354.0053	0.2345	0.008012	0.121360
DLCE ($\hat{s}^F_{u,i}$)	359.7100	0.2345	0.009225	0.121828
Personal Popular (PP)	494.1870	0.2344	0.009441	0.112663

Table 4.8: Comparison on $DH_{personalised}$ with $\mathcal{L}^{Mod}_{PropCare}$

Method	**Avg. Rank Diff.**	**Kendall Tau**	**Top 10 Overlap**	**Top 100 Overlap**
DLCE ($\hat{s}_{u,i}$)	414.7630	0.1269	0.011000	0.060788
DLCE ($\hat{s}^F_{u,i}$)	418.0750	0.1269	0.013339	0.059450
Personal Popular (PP)	542.6320	0.1268	0.012776	0.055071

Due to the availability of more diversified estimates of relevance probability the idea of $R_{u,i}$ evaluation and usage similar to $Z_{u,i}$ through IPS was derived. The intuition is that apart from measuring the causal effect the model should also take into account the treatment effect coming from the item's relevance. However, the abscence of its true values stimulate the metrics collection for various values of ϕ from eq. (13) and the threshold parameter for $Norm(\hat{r}_{u,i}) \geq \epsilon^{'}$. Such simulations were performed only once on validation dataset for different combinations of hyperparameters and their statistics were recorded and reflected in Tables A.1 and A.2 for both settings. This was done as the first step to initialise the trends and select variants yielding best results.

Regarding the former results, the variant with $\phi = 0.1$ and $\epsilon^{'} = 0.5$ should be chosen as it produces higher results in five out of six metrics. It can be noticed that with the increase of $\tilde{\tau}^{'}_{u,i}$ weight in the loss function the performance worsens.

For the chosen hyperparameters in the case of $DH_{original}$ the performance was recorded and its results are represented by Table 4.9. For the DLCE ($\hat{s}_{u,i}$) approach the metrics significantly decreased. However, DLCE ($\hat{s}^F_{u,i}$) yields around the same statistics showing more stable performance overall.

Table 4.9: Metrics with $\mathcal{L}^{Mod}_{PropCare}$ and $\mathcal{L}^{Mod}_{DLCE}$ on $DH_{original}$ (Mean ± Std. Error)

Method	**NDCG@10**	**Recall@10**	**Precision@10**
Relevance (R)	0.0237 ± 0.0144	0.0106 ± 0.0041	0.0159 ± 0.0062
Popularity (P)	0.2223 ± 0.0	0.1202 ± 0.0	0.1755 ± 0.0
Personal Popular (PP)	$\mathbf{0.2477 \pm 0.0}$	$\mathbf{0.1395 \pm 0.0}$	$\mathbf{0.2049 \pm 0.0}$
DLCE ($\hat{s}_{u,i}$)	0.1693 ± 0.0175	0.0961 ± 0.0030	0.1408 ± 0.0044
DLCE ($\hat{s}^{F}_{u,i}$)	0.2436 ± 0.0010	0.1371 ± 0.0004	0.2019 ± 0.0006
DLCE ($\hat{s}^{FI}_{u,i}$)	0.1188 ± 0.0298	0.0693 ± 0.0161	0.1012 ± 0.0242
DLCE ($\hat{s}^{FU}_{u,i}$)	0.0877 ± 0.0226	0.0506 ± 0.0125	0.0771 ± 0.0195

Method	**CP@10**	**CP@100**	**CDCG**
Relevance (R)	0.0071 ± 0.0028	0.0044 ± 0.0008	0.7247 ± 0.0140
Popularity (P)	0.0293 ± 0.0	0.0155 ± 0.0	0.9251 ± 0.0
Personal Popular (PP)	0.0315 ± 0.0	$\mathbf{0.0164 \pm 0.0}$	$\mathbf{0.9396 \pm 0.0}$
DLCE ($\hat{s}_{u,i}$)	0.0223 ± 0.0039	0.0135 ± 0.0001	0.8915 ± 0.0147
DLCE ($\hat{s}^{F}_{u,i}$)	$\mathbf{0.0327 \pm 0.0012}$	0.0162 ± 0.0001	0.9269 ± 0.0033
DLCE ($\hat{s}^{FI}_{u,i}$)	0.0184 ± 0.0066	0.0115 ± 0.0012	0.8413 ± 0.0360
DLCE ($\hat{s}^{FU}_{u,i}$)	0.0159 ± 0.0049	0.0118 ± 0.0012	0.8286 ± 0.0282

The results of the simulations done for $DH_{personalised}$ depict that parameters $\phi = 0.1$ and $\epsilon^{'} = 0.6$ should be picked as they dominate in 3 out of 6 cases. Similar to previous case, parameter ϕ negatively correlates with performance overall.

Simulations results from Table 4.10 are similar to $DH_{original}$ setting in the scope of DLCE ($\hat{s}_{u,i}$) (except for improved CDCG). However, in relation to DLCE ($\hat{s}^{F}_{u,i}$), its results improved noticeably providing with better and closer performance to personal popular.

Table 4.10: Metrics with $\mathcal{L}^{Mod}_{PropCare}$ and $\mathcal{L}^{Mod}_{DLCE}$ on $DH_{personalised}$ (Mean ± Std. Error)

Method	**NDCG@10**	**Recall@10**	**Precision@10**
Relevance (R)	0.0172 ± 0.0014	0.0094 ± 0.0012	0.0154 ± 0.0024
Popularity (P)	0.2316 ± 0.0	0.1150 ± 0.0	0.1870 ± 0.0
Personal Popular (PP)	$\mathbf{0.3431 \pm 0.0}$	$\mathbf{0.1826 \pm 0.0}$	$\mathbf{0.2938 \pm 0.0}$
DLCE ($\hat{s}_{u,i}$)	0.1642 ± 0.0119	0.0964 ± 0.0037	0.1544 ± 0.0063
DLCE ($\hat{s}^{F}_{u,i}$)	0.3028 ± 0.0027	0.1584 ± 0.0011	0.2575 ± 0.0022
DLCE ($\hat{s}^{FI}_{u,i}$)	0.1654 ± 0.0035	0.0884 ± 0.0019	0.1441 ± 0.0030
DLCE ($\hat{s}^{FU}_{u,i}$)	0.1815 ± 0.0129	0.0934 ± 0.0055	0.1552 ± 0.0095

Method	**CP@10**	**CP@100**	**CDCG**
Relevance (R)	0.0084 ± 0.0012	0.0041 ± 0.0005	0.6814 ± 0.0159
Popularity (P)	0.0275 ± 0.0	0.0157 ± 0.0	0.8646 ± 0.0
Personal Popular (PP)	$\mathbf{0.1387 \pm 0.0}$	$\mathbf{0.0392 \pm 0.0}$	$\mathbf{1.4728 \pm 0.0}$
DLCE ($\hat{s}_{u,i}$)	0.0617 ± 0.0037	0.0172 ± 0.0004	0.9648 ± 0.0162
DLCE ($\hat{s}^{F}_{u,i}$)	0.1151 ± 0.0015	0.0336 ± 0.0002	1.3391 ± 0.0034
DLCE ($\hat{s}^{FI}_{u,i}$)	0.0575 ± 0.0023	0.0172 ± 0.0004	0.9607 ± 0.0111
DLCE ($\hat{s}^{FU}_{u,i}$)	0.0727 ± 0.0021	0.0183 ± 0.0006	1.0365 ± 0.0043

Considering the relationship between relevance and studied approaches the statistics from Tables 4.11 and 4.12 show that in comparison with Tables 4.7 and 4.8 the respective difference in lists content increased (based on average rank difference and Kendall tau) but the top-10 overlap seems to improve for both cases and top-100 for $DH_{personalised}$. Such results accompanied by metrics suggest that $\mathcal{L}^{Mod}_{DLCE}$ version of algorithm is not a universal solution for uplift-relevance relation balancing.

Table 4.11: Comparison on $DH_{original}$ with $\mathcal{L}^{Mod}_{PropCare}$ and $\mathcal{L}^{Mod}_{DLCE}$

Method	**Avg. Rank Diff.**	**Kendall Tau**	**Top 10 Overlap**	**Top 100 Overlap**
DLCE ($\hat{s}_{u,i}$)	428.5228	0.0987	0.056431	0.054396
DLCE ($\hat{s}^F_{u,i}$)	435.1803	0.0986	0.021221	0.052239
Personal Popular (PP)	568.9789	0.0985	0.013859	0.040260

Table 4.12: Comparison on $DH_{personalised}$ with $\mathcal{L}^{Mod}_{PropCare}$ and $\mathcal{L}^{Mod}_{DLCE}$

Method	**Avg. Rank Diff.**	**Kendall Tau**	**Top 10 Overlap**	**Top 100 Overlap**
DLCE ($\hat{s}_{u,i}$)	419.0561	0.1114	0.095193	0.116709
DLCE ($\hat{s}^F_{u,i}$)	425.6132	0.1113	0.065743	0.102620
Personal Popular (PP)	529.4819	0.1105	0.033954	0.075639

Final comparison for DH settings was done to differentiate the performance of compared approaches in terms of average price of items recommended and level of their personalisation.

The comparison of average price for top-k items is done based on the results presented at Tables 4.13 and 4.14. Overall, it can be said that personal popular tends to recommend items that have greater price. In addition, the statistical difference between the values obtained from the respective lists produced from DLCE variations was verified with the t-test for all cases except for top-100 case with $\hat{s}_{u,i}$ and $\hat{s}^F_{u,i}$ with $\mathcal{L}^{Mod}_{DLCE}$ on $DH_{personalised}$.

Table 4.13: Price Comparison on $DH_{original}$

Method	**Avg. Price Top-10**	**Avg. Price Top-100**
DLCE ($\hat{s}_{u,i}$) ($\mathcal{L}_{DLCE}$)	2.5052 ± 0.4057	2.6563 ± 0.1025
DLCE ($\hat{s}^F_{u,i}$) ($\mathcal{L}_{DLCE}$)	2.5822 ± 0.4676	2.6892 ± 0.1223
DLCE ($\hat{s}_{u,i}$) ($\mathcal{L}^{Mod}_{DLCE}$)	2.5188 ± 0.3945	2.6904 ± 0.1184
DLCE ($\hat{s}^F_{u,i}$) ($\mathcal{L}^{Mod}_{DLCE}$)	2.5980 ± 0.4906	2.7350 ± 0.1417
Personal Popular (PP)	2.6257 ± 0.5204	2.8584 ± 0.2056

Table 4.14: Price Comparison on $DH_{personalised}$

Method	**Avg. Price Top-10**	**Avg. Price Top-100**
DLCE ($\hat{s}_{u,i}$) ($\mathcal{L}_{DLCE}$)	2.6172 ± 0.4908	2.6419 ± 0.1346
DLCE ($\hat{s}^F_{u,i}$) ($\mathcal{L}_{DLCE}$)	2.5351 ± 0.4895	2.6551 ± 0.1380
DLCE ($\hat{s}_{u,i}$)($\mathcal{L}^{Mod}_{DLCE}$)	2.7478 ± 0.5435	2.7659 ± 0.1511
DLCE ($\hat{s}^F_{u,i}$) ($\mathcal{L}^{Mod}_{DLCE}$)	2.6229 ± 0.5312	2.7628 ± 0.1614
Personal Popular (PP)	2.5991 ± 0.5645	2.8618 ± 0.2027

Algorithms were also tested based on how personalised they are are using Jaccard score with values close to one meaning more unique recommendations for users. Table 4.15 suggests that on average DLCE ($\hat{s}^F_{u,i}$) with either version of loss function outperforms other approaches.

Table 4.15: Personalization Score Comparison

Method	$DH_{original}$	$DH_{personalised}$
DLCE ($\hat{s}_{u,i}$) ($\mathcal{L}_{DLCE}$)	0.820	0.943
DLCE ($\hat{s}^F_{u,i}$) ($\mathcal{L}_{DLCE}$)	0.870	0.912
DLCE ($\hat{s}_{u,i}$) ($\mathcal{L}^{Mod}_{DLCE}$)	0.769	0.928
DLCE ($\hat{s}^F_{u,i}$) ($\mathcal{L}^{Mod}_{DLCE}$)	0.877	0.912
Personal Popular (PP)	0.879	0.910

As a final part of the research it was decided to perform simulation of algorithms on non-synthesised dataset FINN.no. This was done to verify the positive influence of frequency incoporation into prediction of DLCE as well as too confirm whether or not it is capable to outperform personal popular approach.

An important remark should be made considering the specifity of the utilised dataset. To begin with, it must be mentioned that FINN.no is considerably smaller than both versions of DunnHumby. In addition, it is quite sparse making the popularity scores concentrate around 0 (Figure 4.3 depicts min-max scaled values distribution).

Both points result in inability of DLCE to properly train resulting in performance worse than randomization (supported by the Figure A.1). The popularity approach in its nature resembles the randomization and is perceived in this formulation. However, the goal of FINN.no utilisation is to rather focus on the comparison of DLCE and PP performance in a different setting.

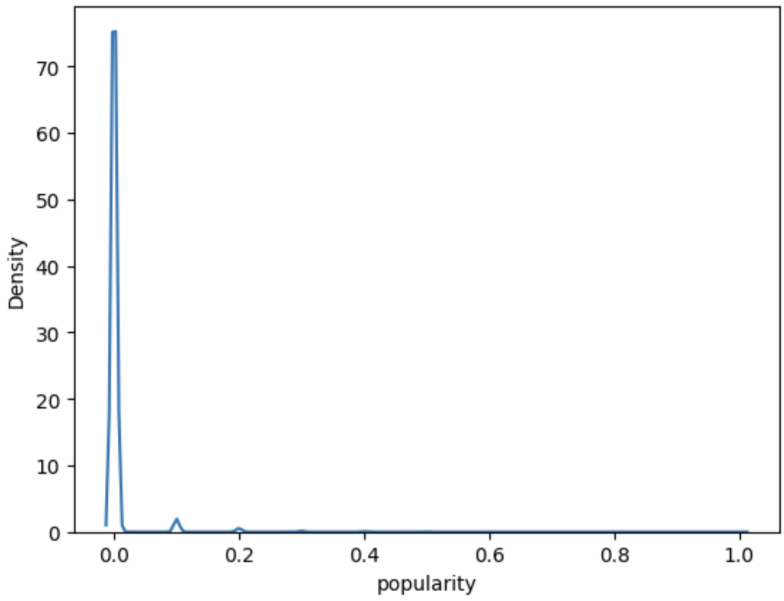

Figure 4.3: Distribution of popularity in FINN.no dataset

Taking into consideration mentioned reasoning, the analysis of simulations, that were performed 10 consecutive times (to mitigate the effect of datasets size), begins with the approaches results depicted in Table 4.16.

Table 4.16: Results of Metrics on FINN.no

Method	**NDCG@10**	**Recall@10**	**Precision@10**	**Weighted Average Uplift**
DLCE ($\hat{s}_{u,i}$)	0.2468 ± 0.0093	0.4165 ± 0.0104	0.0982 ± 0.0020	-0.0475
DLCE ($\hat{s}^{F}_{u,i}$)	0.2194 ± 0.0014	0.4028 ± 0.0022	0.0937 ± 0.0005	-0.0465
DLCE ($\hat{s}^{FI}_{u,i}$)	0.2626 ± 0.0041	0.4372 ± 0.0055	0.1032 ± 0.0009	-0.0473
DLCE ($\hat{s}^{FU}_{u,i}$)	0.2521 ± 0.0084	0.4284 ± 0.0116	0.0996 ± 0.0027	-0.0474
Personal Popular (PP)	0.2193 ± 0.0019	0.4024 ± 0.0029	0.0936 ± 0.0006	**-0.0464**
Popularity (P)	$\mathbf{0.2709 \pm 0.0043}$	$\mathbf{0.4380 \pm 0.0088}$	$\mathbf{0.1036 \pm 0.0019}$	-0.0480

The analysis of standard metrics allows to make several conclusions. First and foremost, even though popularity outperforms all of the approaches its standard deviation is one of the heighest across all statistics. This reinforces the idea to perceive it as randomized output. Secondly, it can be observed that DLCE ($\hat{s}^{F}_{u,i}$) has close but slightly higher metrics than PP in the current setting. In addition, its deviation is the smallest compared to others which reaffirmes in the assumption that addition of frequency to model's prediction makes its performance more stable. Finally, across DLCE approaches the ($\hat{s}^{FI}_{u,i}$) modification provides with the best metrics.

Considering the causal metric of weighted uplift, PP tends to provide with the best average performance followed closely by DLCE ($\hat{s}^{F}_{u,i}$). This could signal that in the long-run provision of more data could have improved the results. To provide with the analysis based on the current setting the Uplift@k metric was calculated and visualised which is depicted on Figure 4.4.

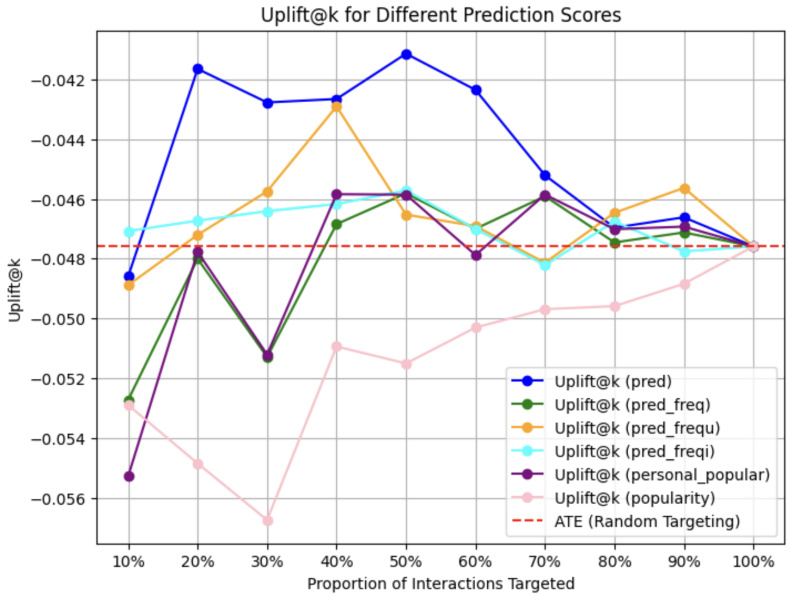

Figure 4.4: Distribution of popularity in FINN.no dataset

Based on the provided statistics popularity approach seems to perform the worst among all. In addition, Most of DLCE approaches tend to provide better results than the randomisation. What is interesting in this case is that the DLCE ($\hat{s}^{FI}_{u,i}$) method which provides best metrics, excluding popularity, seems to converge for a random performance more than others do. Thus, it cannot be fully suggested that this approach would definitely outperform DLCE ($\hat{s}^{F}_{u,i}$) version provided more data. Lastly, a strong relationship between PP and DLCE ($\hat{s}^{F}_{u,i}$) can be observed with them alternately changing leadership.

5 Conclusions and future prospects

The current paper presents the idea behind the uplift modeling setting and provides with an evaluation of potential modifications that can be made for the purpose of attempts to account for the effect of item's relevance on user's interaction with it.

The description of obtained results in both DunnHumby settings suggests that usage of DLCE ($\hat{s}^{F}_{u,i}$) should be prioritised. First of all, its performance in comparison with the DLCE ($\hat{s}_{u,i}$) is noticeably better and is more stable in various simulation settings. In addition, in the case of $\mathcal{L}^{Mod}_{DLCE}$ implementation, the derived method shows non-worse results almost for all metrics for $DH_{original}$ setting compared to initial realisation and substantially enhanced onced for $DH_{personalised}$. Considering its relationship with relevance, DLCE ($\hat{s}^{F}_{u,i}$) allows one to provide around the same or even better statistics compared to DLCE ($\hat{s}_{u,i}$) and substantially better with regard to personal popular approach. Furthermore, the discussed method also excels in the pro-

vision of more individual recommendations.

Acknowledging the limited scope of FINN.no setting, the analysis performed on it still allowed to derive some useful considerations. The received metrics indicate more stable long-term potential performance, suggesting as well that personal popular approach is not the best fit for all cases.

However, an important assumption is made in the research of the availability of user-item frequency interaction data. As a result, the derived approach is subject to cold start problem due to lack of such available information for new users or items.

Considering the future prospects of the work, several ideas could be realised and tested. Firstly, the research would benefit to be complemented with a large non-synthetic dataset which would allow to make more definite considerations. Secondly, the solutions for the mitigation of cold start problem could be tested with the leverage of popularity intial usage for new users and other items features used. In addition, the new ideas for creation of a more versatile and much better performing approach that incorporates the item's relevance should probably be considered.

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A Appendix

Algorithm 1 PropCare and DLCE Training

Input: Observed training data D, η, γ, ω, χ^T, χ^C
Output: Weights DLCE (θ_{DLCE})

Initialize PropCare weights: θ_{PropCare}
for $e = 1$ to E **do**
  for pair (u, i) in D **do**
    Compute $\mathcal{L}_{\text{naive}}$ from eq. (4)
    Sample an item j for each pair from $I \setminus \{i\}$
    Compute $\mathcal{L}_{\text{pop}}$ from eq. (5)
    Compute $\mathcal{L}_{\text{PropCare}}$ from eq. (6)
  end for
  Update θ_{PropCare} with backpropagation
end for
Obtain exposure data from $\theta_{PropCare}$
Initialize DLCE weights: θ_{DLCE}
for $n = 1$ to N **do**
  Sample a pair (u, i) from D
  Sample an item j for each pair from $I \setminus \{i\}$
  if $Z_{u,i} = 1$ **then**
    $\theta_{DLCE} = \theta_{DLCE} + \eta(-\frac{1}{\max(p_{u,i}, \chi^T)} \frac{\partial}{\partial t}(\log(1 + \exp(\omega \hat{s}_{u,i,j}))) - \gamma \theta_{DLCE})$
  else
    $\theta_{DLCE} = \theta_{DLCE} + \eta(-\frac{1}{\max((1 - p_{u,i}), \chi^C)} \frac{\partial}{\partial t}(\log(1 + \exp(-\omega \hat{s}_{u,i,j}))) - \gamma \theta_{DLCE})$
  end if
end for

Return: θ_{DLCE}

Table A.1: Metrics for DLCE ($\hat{s}_{u,i}$) with $\mathcal{L}_{DLCE}^{Mod}$ on $DH_{original}$

ϕ	**Threshold**	**NDCG@10**	**Recall@10**	**Precision@10**
0.1	0.5	**0.1398**	**0.0911**	**0.1341**
0.2	0.5	0.1190	0.0744	0.1089
0.3	0.5	0.1118	0.0687	0.1013
0.1	0.6	0.1384	0.0901	0.1319
0.2	0.6	0.1154	0.0694	0.1036
0.3	0.6	0.1115	0.0678	0.1000
0.1	0.7	0.1329	0.0851	0.1258
0.2	0.7	0.1150	0.0699	0.1036
0.3	0.7	0.1095	0.0664	0.0983

ϕ	**Threshold**	**CP@10**	**CP@100**	**CDCG**
0.1	0.5	**0.0239**	0.0129	**0.8584**
0.2	0.5	0.0222	0.0118	0.8354
0.3	0.5	0.0220	0.0120	0.8346
0.1	0.6	0.0224	0.0128	0.8458
0.2	0.6	0.0217	0.0124	0.8485
0.3	0.6	0.0223	0.0123	0.8503
0.1	0.7	0.0213	0.0129	0.8531
0.2	0.7	0.0224	**0.0130**	0.8581
0.3	0.7	0.0232	0.0121	0.8438

Table A.2: Metrics for DLCE ($\hat{s}_{u,i}$) with $\mathcal{L}_{DLCE}^{Mod}$ on $DH_{personalised}$

ϕ	**Threshold**	**NDCG@10**	**Recall@10**	**Precision@10**
0.1	0.5	0.1834	0.1010	**0.1655**
0.2	0.5	0.1755	0.0954	0.1558
0.3	0.5	0.1693	0.0886	0.1461
0.1	0.6	**0.1843**	0.1011	0.1647
0.2	0.6	0.1740	0.0938	0.1547
0.3	0.6	0.1685	0.0881	0.1465
0.1	0.7	0.1831	**0.1015**	0.1654
0.2	0.7	0.1735	0.0938	0.1548
0.3	0.7	0.1677	0.0891	0.1469

ϕ	**Threshold**	**CP@10**	**CP@100**	**CDCG**
0.1	0.5	0.0615	0.0197	1.0317
0.2	0.5	0.0554	0.0189	1.0066
0.3	0.5	0.0507	0.0187	1.0025
0.1	0.6	0.0613	**0.0197**	**1.0357**
0.2	0.6	0.0551	0.0180	1.0032
0.3	0.6	0.0515	0.0181	0.9911
0.1	0.7	**0.0623**	0.0193	1.0315
0.2	0.7	0.0542	0.0187	1.0040
0.3	0.7	0.0511	0.0188	0.9873

Table A.3: Mean Results for $DH_{original}$

Category	Method	CP@10	CP@100	CDCG	NDCG@10	Recall@10	Precision@10
Baselines	Popularity	0.0293	0.0155	0.9251	0.2223	0.1202	0.1755
	Personal Popularity	0.0315	0.0164	0.9396	0.2477	0.1395	0.2049
$\mathcal{L}_{\mathbf{PropCare}}$ **and** $\mathcal{L}_{\mathbf{DLCE}}$	Relevance PropCare	0.0022	0.0024	0.6677	0.0014	0.0010	0.0013
	DLCE ($\hat{s}_{u,i}$)	0.0284	0.0147	0.8928	0.1952	0.1106	0.1631
	DLCE ($\hat{s}^{F}_{u,i}$)	**0.0349**	0.0158	0.9187	0.2434	0.1365	0.2011
	DLCE ($\hat{s}^{FI}_{u,i}$)	0.0167	0.0112	0.8185	0.1148	0.0678	0.0981
	DLCE ($\hat{s}^{FU}_{u,i}$)	0.0155	0.0108	0.8155	0.0960	0.0545	0.0826
$\mathcal{L}^{\mathbf{Mod}}_{\mathbf{PropCare}}$ **and** $\mathcal{L}_{\mathbf{DLCE}}$	Relevance PropCare	0.0130	0.0045	0.7492	0.0117	0.0072	0.0109
	DLCE ($\hat{s}_{u,i}$)	0.0289	0.0143	0.9042	0.1982	0.1097	0.1618
	DLCE ($\hat{s}^{F}_{u,i}$)	0.0328	0.0161	0.9254	**0.2455**	**0.1381**	**0.2035**
	DLCE ($\hat{s}^{FI}_{u,i}$)	0.0230	0.0133	0.8644	0.1533	0.0887	0.1292
	DLCE ($\hat{s}^{FU}_{u,i}$)	0.0197	0.0126	0.8475	0.1272	0.0741	0.1112
$\mathcal{L}^{\mathbf{Mod}}_{\mathbf{PropCare}}$ **and** $\mathcal{L}^{\mathbf{Mod}}_{\mathbf{DLCE}}$	Relevance PropCare	0.0071	0.0044	0.7247	0.0237	0.0106	0.0159
	DLCE ($\hat{s}_{u,i}$)	0.0223	0.0135	0.8915	0.1693	0.0961	0.1408
	DLCE ($\hat{s}^{F}_{u,i}$)	0.0327	**0.0162**	**0.9269**	0.2436	0.1371	0.2019
	DLCE ($\hat{s}^{FI}_{u,i}$)	0.0184	0.0115	0.8413	0.1188	0.0693	0.1012
	DLCE ($\hat{s}^{FU}_{u,i}$)	0.0159	0.0118	0.8286	0.0877	0.0506	0.0771

Bold values indicate the best DLCE Ranking performance for each metric.

Table A.4: Mean Results for $DH_{personalised}$

Category	Method	CP@10	CP@100	CDCG	NDCG@10	Recall@10	Precision@10
Baselines	Popularity	0.0275	0.0157	0.8646	0.2316	0.1150	0.1870
	Personal Popularity	0.1387	0.0392	1.4728	0.3431	0.1826	0.2938
$\mathcal{L}_{\mathbf{PropCare}}$ **and** $\mathcal{L}_{\mathbf{DLCE}}$	Relevance PropCare	0.0015	0.0019	0.6128	0.0012	0.0008	0.0012
	DLCE ($\hat{s}_{u,i}$)	0.0569	0.0172	0.9438	0.1816	0.1052	0.1699
	DLCE ($\hat{s}^{F}_{u,i}$)	0.0959	0.0271	1.2195	0.2892	0.1495	0.2449
	DLCE ($\hat{s}^{FI}_{u,i}$)	0.0407	0.0127	0.8616	0.1400	0.0764	0.1228
	DLCE ($\hat{s}^{FU}_{u,i}$)	0.0657	0.0144	0.9889	0.1602	0.0819	0.1361
$\mathcal{L}^{\mathbf{Mod}}_{\mathbf{PropCare}}$ **and** $\mathcal{L}_{\mathbf{DLCE}}$	Relevance PropCare	0.0049	0.0044	0.6613	0.0145	0.0090	0.0146
	DLCE ($\hat{s}_{u,i}$)	0.0591	0.0185	0.9572	0.1800	0.1042	0.1686
	DLCE ($\hat{s}^{F}_{u,i}$)	0.1030	0.0294	1.2695	0.2946	0.1513	0.2480
	DLCE ($\hat{s}^{FI}_{u,i}$)	0.0513	0.0165	0.9302	0.1582	0.0859	0.1377
	DLCE ($\hat{s}^{FU}_{u,i}$)	0.0700	0.0175	1.0161	0.1843	0.0990	0.1626
$\mathcal{L}^{\mathbf{Mod}}_{\mathbf{PropCare}}$ **and** $\mathcal{L}^{\mathbf{Mod}}_{\mathbf{DLCE}}$	Relevance PropCare	0.0084	0.0041	0.6814	0.0172	0.0094	0.0154
	DLCE ($\hat{s}_{u,i}$)	0.0617	0.0172	0.9648	0.1642	0.0964	0.1544
	DLCE ($\hat{s}^{F}_{u,i}$)	**0.1151**	**0.0336**	**1.3391**	**0.3028**	**0.1584**	**0.2575**
	DLCE ($\hat{s}^{FI}_{u,i}$)	0.0575	0.0172	0.9607	0.1654	0.0884	0.1441
	DLCE ($\hat{s}^{FU}_{u,i}$)	0.0727	0.0183	1.0365	0.1815	0.0934	0.1552

Bold values indicate the best DLCE Ranking performance for each metric.

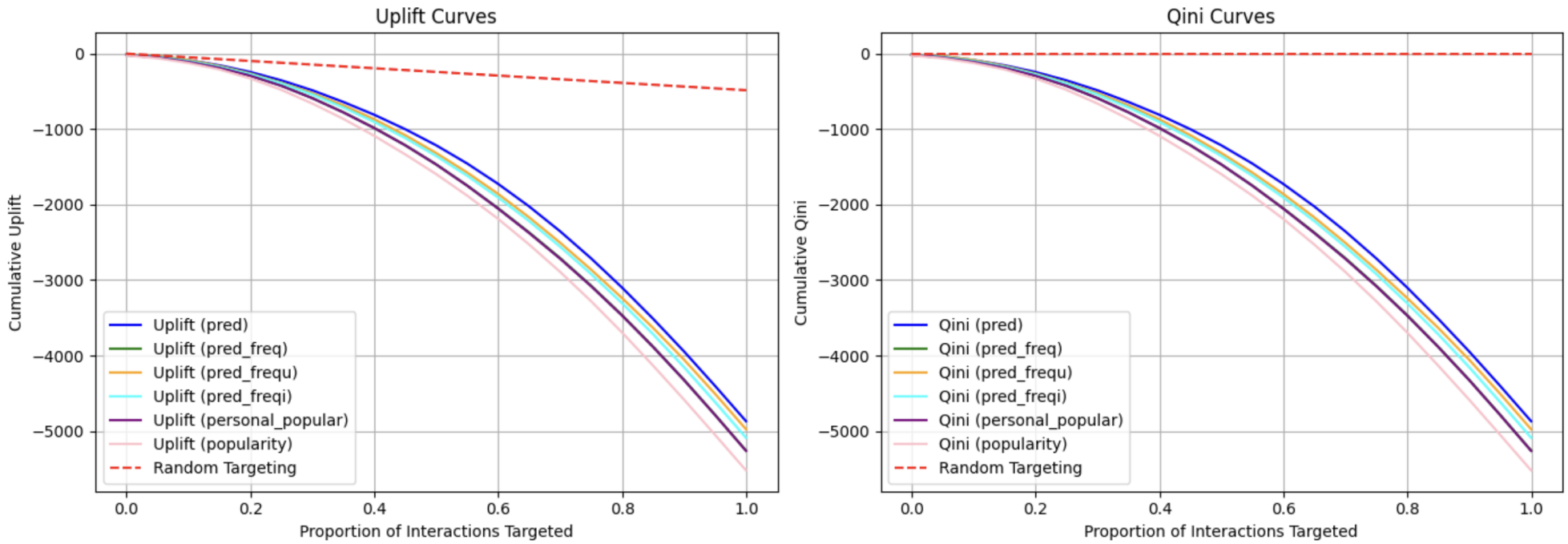

Figure A.1: Uplift and Qini Curves on FINN.no dataset